

100 GHz Ge/Si Avalanche Photodiodes Enabling Net 400 Gbps Data Reception at a Low Voltage of 5 V

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Abstract We present Ge/Si avalanche photodiodes operating at low voltages (<6.5 V) achieving BWs of 100 GHz and 50 GHz at 1.8 A/W and 3.7 A/W responsivity levels across the O- and C-bands, enabling net 400 Gbps PAM-4 data link demonstration below the FEC threshold.

Introduction

In the era of hyper-scale data centers, artificial intelligence (AI) workloads, and the rapid rollout of 5G/6G networks, the global demand for data bandwidth (BW) is expanding at an exponential rate, putting significant pressure on the entire optical interconnect infrastructure across all domains. As the industry shifts towards aggregate data rates of 3.2 Tbps and 6.4 Tbps for scale-up and scale-out connectivity inside the data centres [1], and to data rates of 100 Gbps and 200 Gbps for very high speed passive optical networks (VHSP) [2], maintaining the same reach and power budget for those high speed links has become a critical challenge [3]. On the receiver side of those links, Ge/Si Avalanche Photodiodes (APDs) have the advantage of increasing the receiver sensitivity, helping to close the link budget or increase the link reach, while being compatible with complementary metal oxide semiconductor (CMOS) fabrication processes, enabling high density integration [4]. Ge/Si APDs have been demonstrated in either intensity modulation direct detection (IMDD) links improving the receiver sensitivity by 5 dB to 8 dB [5-7], or in coherent links lowering the required local oscillator (LO) power [8]. Previously in [9,10], we showcased a separate absorption, charge and multiplication (SACM) Ge/Si APD fabricated on imec's 300 mm silicon photonics platform [11], achieving a BW of 70 GHz at 2 A/W responsivity at an operating voltage of -7 V. In this work, we improve the performance of our APD by using a thin multiplication layer of 100 nm with no charge layer, along with high Si slab doping. This enabled the device breakdown voltage to be reduced from -9.5 V to -6.5 V while pushing its maximum BW to more than 110 GHz (measurement instrument limit) at unity gain (responsivity of 0.95 A/W). The device maintains BWs of 100 GHz and 50 GHz at responsivity levels of 1.8 A/W and 3.9 A/W, respectively. Finally, we demonstrate data reception at

1550 nm with a net rate of 400 Gbps (425 Gbps gross) using 212.5 Gbaud PAM-4, achieving performance below the 6.25% overhead forward error correction (6.25% OH-FEC) threshold.

Device Design and Structure

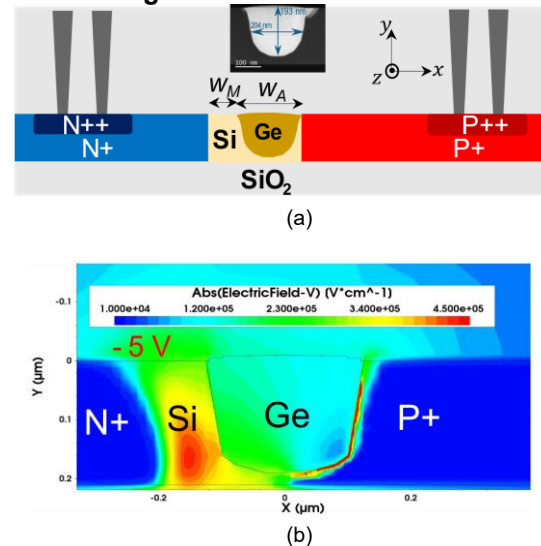


Fig. 1: (a) Cross-section view representing the lateral Ge/Si SAM APD with an inset TEM image of the Ge absorption region recessed inside the Si slab. (b) TCAD simulation of electric field profile at a voltage bias of -5 V.

The APD presented in this work is a lateral Ge/Si waveguide coupled APD implemented using the separate absorption and multiplication structure (SAM). Fig. 1 (a) shows the cross section of the reported device where a narrow Ge absorption region of width, $W_A = 200$ nm, is deeply recessed inside the Si slab and an intrinsic Si region of width, $W_M = 100$ nm, acts as the multiplication region. The inset in Fig. 1 (a) shows a transmission electron microscopy (TEM) image of the deeply recessed Ge region having a width of 204 nm and a height of 193 nm. The Si slab doping levels, N+ and P+ are in the order of $2 \times 10^{19} \text{ cm}^{-3}$. These high doping levels combined with

the narrow multiplication region of 100 nm enhanced the magnitude of the electric field profile inside the Si multiplication region as shown in the technology computer aided design (TCAD) simulation, Fig. 1 (b). At a relatively low voltage of -5 V the electric field reaches a value of more than 450 kVcm^{-1} inside the Si multiplication region which is sufficient for a gain of 2. By removing the charge region, the device fabrication process is simplified as implementing a narrow charge region with high uniformity across the wafer is challenging. Usually, the charge region is used to protect the Ge region from the high electric field value in the multiplication region which could cause impact ionization to happen in Ge degrading the APD over-all performance. This is generally true, but for this structure combination of thin Si multiplication region of width 100 nm and Ge absorption region of width 200 nm and for low gain values of less than 6, the electric field reduction caused by the dielectric constant difference across the Ge/Si interface along with trapped charges at the Ge/Si interface, reported by different groups [12-14], helps to keep the electric field inside the Ge below the impact ionization threshold.

Device Performance

First, the DC performance is characterized at both wavelengths of 1310 nm and 1550 nm. The APDs used for the two wavelengths only differ in their length, where the O-band APD has a length of $10 \mu\text{m}$, while the C-band APD has a length of $30 \mu\text{m}$. The lengths of both devices were chosen to achieve the highest primary responsivity at the respective operating wavelength. Fig. 2 shows the measured IV curves for 12 selected dies at both wavelengths along with the corresponding responsivity curves. The responsivity is calculated as $(I_{\text{light}} - I_{\text{dark}})/P_{\text{in}}$, where P_{in} is the optical power reaching the device, taking the grating coupler loss into account. An input optical power of -10 dBm in the device was used. Both devices show a breakdown voltage around -6.5 V with the

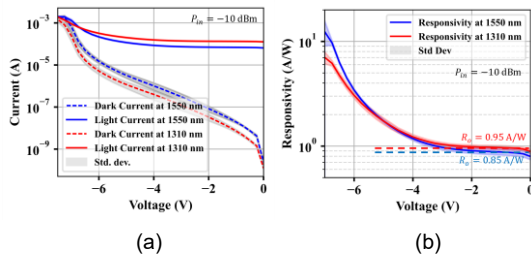


Fig. 2: (a) Dark and light current measured at an input optical power of -10 dBm in the waveguide for 12 selected dies at wavelengths of 1310 nm and 1550 nm. (b) Extracted responsivity of the Ge/Si APDs for 12 selected dies at wavelengths of 1310 nm and 1550 nm.

dark current reaching a value of $10 \mu\text{A}$ at around -6 V. The C-band APD has a higher dark current than the O-band one, which is expected since the C-band APD is longer. As for the responsivity, it reaches a value of around 2 A/W at -5.5 V and 4 A/W at -6.5 V for both devices. For the measured 12 dies, both devices show good performance repeatability. Next, the small signal RF performance was characterized on 12 dies using the same setup in [9]. Fig. 3 shows the measured opto-electrical (OE) response at an input optical power of -10 dBm in the waveguide, along with the extracted BW vs. voltage and BW vs. gain at both wavelengths of 1310 nm and 1550 nm. For both devices the BW exceeds the maximum measurement setup limitation of 110 GHz starting from a voltage of -1 V.

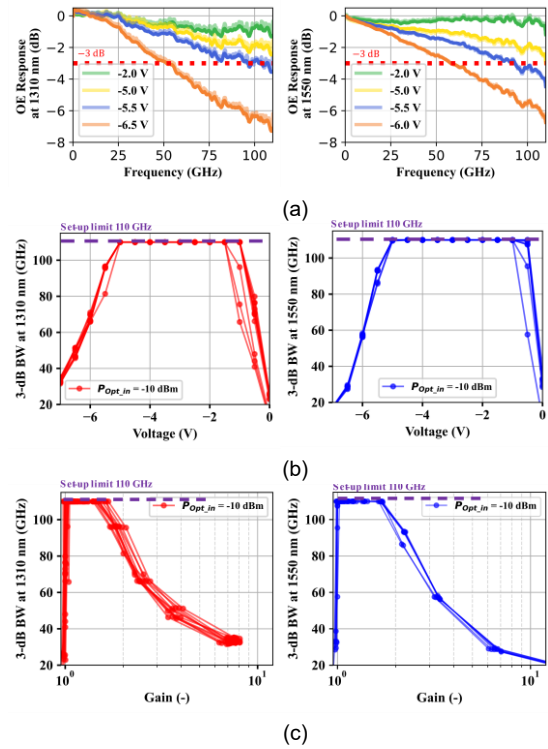


Fig. 3: Small signal RF performance of the Ge/Si APDs measured at the wavelengths of 1310 nm and 1550 nm. (a) Normalized OE response for one device at each wavelength. (b) BW vs Voltage for 12 devices at each wavelength. (c) BW vs Gain for 12 devices at each wavelength.

It can be noted from the OE curves, that at -2 V a reduction of less than 1.5 dB is observed at 110 GHz for both devices indicating that the real BW is much higher. As the reverse bias is increased, the BW drops to a value of 100 GHz at a reverse bias of -5 V, for both devices, while it drops to 50 GHz at a reverse bias of -6.5 V for the O-band device and -6 V for the C-band one. At 100 GHz BW, the mean gain is around 1.85 and 2 for the O- and C-band APDs, respectively. This corresponds to a responsivity of around 1.8 A/W for both devices. The mean responsivity increases to

around 3.7 A/W for the O-band device and 3.5 A/W for the C-band one at a BW of 50 GHz. It is also interesting to note that, although both devices show good repeatability across the 12 dies in the BW vs. voltage curves, the O-band devices show around 15 % variation in the BW vs. gain curves while the C-band devices show better repeatability. This could be attributed to the O-band devices being more sensitive to the space charge effect [15], causing gain variations. More investigation is needed to better understand this behaviour.

High Speed Link Demonstration

Finally, to assess the potential of the reported APD for use in optical links targeting 400 Gbps/lane, the C-band APD was tested with a high-speed IM/DD setup. The setup consisted of an arbitrary waveform generator (AWG; M8199B, Keysight Technologies) generating a PAM4 signal from a 15-bit pseudo-random bit stream (PRBS15). The signal was shaped using a root raised cosine (RRC) filter with a 0.05 roll-off factor and a 21-tap FIR filter was used to pre-emphasize the signal compensating for the electric BW limitations of the setup, Fig. 4 (c). The generated signal then drove a 110 GHz waveguide coupled Ge/Si electro-absorption modulator (EAM) developed by imec [16]. A laser source with a wavelength of 1550 nm was used as the optical carrier feeding the EAM. After the EAM, an erbium-doped fiber amplifier (EDFA; CEFA-C-PB-HP, Keopsys) was employed to compensate for the setup losses, which are mainly dominated by the grating couplers feeding the input/output optical signal of the EAM and the input to the APD. The combined losses from the grating couplers are around 15 dB. The APD is connected to a 110 GHz sampling scope (DCA-X, Keysight Technologies). Both EAM and APD devices are probed on-wafer using 110 GHz GSG RF probes and DC bias is supplied through 1-mm bias-tees. Short coaxial cables are used for the interconnects. After capturing the signal, offline digital signal processing (DSP) with a T/2-spaced 300-Tap FIR equalization filter is employed to compensate for remaining signal distortion, mainly dominated by the limited bandwidth of the used arbitrary waveform generator. Fig. 4 (a) and (b) show the measured PAM-4 signal at data rates of 425 Gbps and 448 Gbps. The measurements were performed at a reverse bias of 5 V, corresponding to a responsivity of around 1.8 A/W at a wavelength of 1550 nm. The photocurrent passing through the APD was around 2.5 mA, which corresponds to an optical input power, at the device, of around 1.4 dBm. The corresponding bit error rate (BER) at the two data rates is shown in Fig.

5 (d). At 425 Gbps the BER is below the 6.25% FEC threshold of 4.5×10^{-3} and is marginally at the 12% overhead FEC of 2×10^{-2} at 448 Gbps. This means a net 400 Gbps reception is achievable using this APD. The reported BER values are mainly limited by the electrically limited setup BW and the sampling scope noise floor. As seen in Fig. 4 (c), the full link electrical response (S21) is mainly dominated by the electrical back-to-back (EB2B) link response of the AWG and the sampling scope. As a reference, the BER values measured for the EB2B link after applying the same DSP routine are plotted in Fig. 4 (d), highlighting the limitation of the measurement setup.

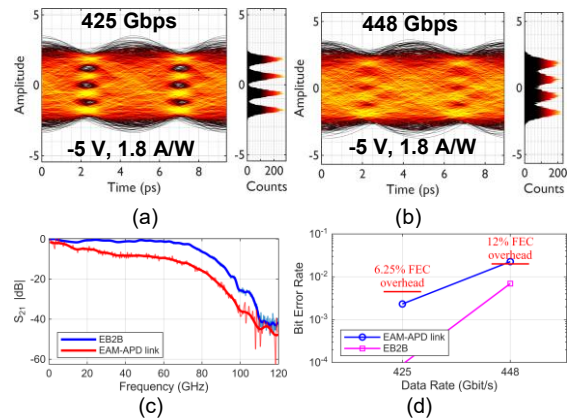


Fig. 4: (a) Measured eye diagram at a gross data rate of 425 Gbps (212.5 Gbaud PAM4) using the Ge/Si APD at a wavelength of 1550 nm. (b) Measured eye diagram at a gross data rate of 448 Gbps (224 Gbaud PAM4) using the Ge/Si APD at a wavelength of 1550. (c) measured electrical response (S21) of the electrical B2B link of the AWG directly connected to the sampling scope and the S21 response of the entire link including the AWG, EAM, APD, sampling scope, RF cables, bias-tees and connectors. (d) BER values measured at gross data rates of 425 Gbps and 448 Gbps for the entire link and the electrical B2B connection.

Conclusion

In this work, we showcased a waveguide Ge/Si APD with a record-high BW exceeding 110 GHz at unity gain and maintaining a BW of 100 GHz at a responsivity of 1.8 A/W across the O- and C-bands, and a BW of 50 GHz at a responsivity of 3.7 A/W across both bands. This performance is achieved at relatively low operating voltages of less than 6.5 V further improving the practical adoption of those devices. A data link demonstration of gross data rates up to 448 Gbps was also carried out, showing the strong potential of those devices to be deployed in next generation short reach optical links targeting 400 Gbps/lane operation. As a next step, these devices need to be integrated with high-speed TIAs to quantitatively assess the receiver sensitivity improvement they could provide.

Acknowledgements

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